Cost Effective Housing Techniques

Unit III: Alternative Building Materials and Low Cost Infrastructure

Syllabus

Alternative building materials for low cost housing: introduction - substitute for scarce materials - ferro-cement - gypsum boards - timber substitutions - industrial wastes - agricultural wastes - alternative building maintenance. Low cost infrastructure services: introduction - present status - technological options - low cost sanitation - domestic waste - water supply - energy.

1. Introduction

Conventional materials like **cement, steel, burnt bricks, timber and sand** are becoming **scarce and costly** and their production consumes much **energy** and damages the environment (top soil loss, forest cutting, CO_2). So **alternative building materials** made from **local, renewable, waste** resources are needed to cut the cost of housing and protect the environment.

Desirable properties of an alternative material

* **Cheap** and **locally available**, easy to produce by simple tools and local skill
* Adequate **strength, durability, fire and weather resistance**
* **Low embodied energy** (energy used in making) and eco-friendly
* **Acceptable** to the people; standardised; Indian Standards (BIS) or **BMTPC PACS** appraisal

Material	Embodied energy (MJ/kg, approx.)
Steel	25 - 35
Aluminium	150-200
Cement	4 - 5
Burnt clay brick	2 - 3
Concrete (plain)	1 - 1.5
Fly ash brick	0.5 - 1
Stabilised soil block	0.7 - 1
Timber (local)	0.5 - 2
Bamboo, mud, straw	Very low (0.1 - 0.5)

2. Substitutes for Scarce Materials

Scarce material	Substitutes
Cement	**Lime-pozzolana (lime + fly ash / surkhi)**, **Portland pozzolana cement (PPC)** with fly ash, **slag cement (PSC)**, rice husk ash cement, **magnesium oxychloride (Sorel) cement**, **lime-fly ash-gypsum**, clay-based mortars, **alkali-activated (geopolymer)** binders
Steel	**Bamboo reinforcement**, **ferrocement** (wire mesh), **prestressed concrete**, **HYSD / TMT bars (less quantity)**, **brick reinforcement**, GFRP / basalt bars, plastic mesh
Timber	**Bamboo, RCC / ferrocement, steel, aluminium, FRP, PVC, particle board**, see section 5
Burnt clay bricks	**Fly ash bricks, FaL-G, sand-lime bricks, stabilised soil blocks, concrete blocks, AAC blocks, stone**, slag bricks
River sand	**Manufactured sand (M-sand)** from crushed stone, **quarry dust, fly ash, crushed recycled aggregate**
Coarse aggregate	**Slag, clinker, recycled C&D; waste, light weight aggregate (fly-ash aggregate, expanded clay)**
Roofing sheets / tiles	**Ferrocement / micro-concrete tiles (MCR), fibre-cement sheets, bamboo mat corrugated sheets, thatch** with treatment, recycled plastic
Plaster	**Gypsum plaster, mud-lime plaster, cement-fly ash mortar**

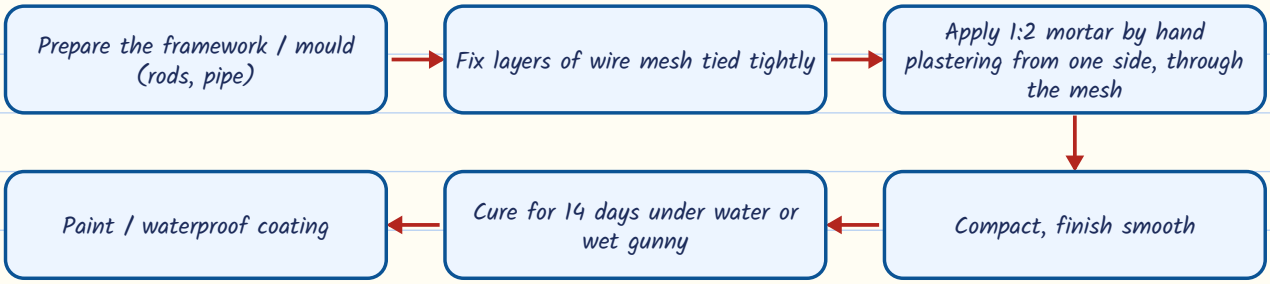
Lime and lime-pozzolana

- **Lime** (from limestone / kankar) is an old binder; **hydraulic lime** sets under water; **fat lime + pozzolana** (fly ash, surkhi (burnt-brick powder), rice husk ash, calcined clay) gives strength slowly, **cheaper than cement and consumes less energy**
- **Lime-pozzolana mortar 1:2** for masonry and plaster, **lime concrete** for foundations and roof terracing (**lime terracing** with brick-bats, jaggery, fenugreek, **Madras terrace**) - traditional, durable, breathable
- Also good for **repair of old buildings and heritage** structures

3. Ferro-Cement

Ferrocement is a thin composite of **rich cement-sand mortar** reinforced with **closely spaced, multiple layers of fine wire mesh** (and sometimes small-diameter rods). The thickness is only **10 – 40 mm**; and it behaves like homogenous elastic material with very high **crack control and impact resistance**.

Component	Details
Mortar	Cement : sand = **1 : 1.5 to 1 : 2.5**; **w/c 0.4 - 0.45**; sand passing 2.36 mm, fine; with **plasticiser**; strength M30 - M40
Wire mesh	**Welded mesh or woven (chicken) mesh, hexagonal**, gauge 20-24, opening 6-25 mm, galvanised; **2-6 layers**
Skeletal steel	Rods 4-6 mm @ 50-100 mm to form framework
Volume fraction of steel	**2 - 8%** (about 300 - 500 kg per m^3)
Thickness	**10 - 40 mm** (covers 2-3 mm)
Curing	Water cure **7 - 14 days** (very important, thin section)



- **Applications in housing: roofing elements (channels, folded plates, shells and domes), wall panels and partitions, water tanks (storage), grain bins (silos), septic tanks, manholes, sunshades, door and window frames, shutters, lintels, latrine units, bio-gas holders, boats, furniture, pipes, precast ferrocement roofing units - "channel roof"**
- **Advantages:** very **thin and light** (low dead load), **cheap** with local skill, **no formwork** (or little), tough and **crack resistant**, easy to repair, **fire and rot proof**, **can take any shape**, 30-50% cheaper than RCC for thin members
- **Limitations: labour intensive**, **corrosion** of the mesh (needs good cover and treatment), difficult to **join and fasten**, **local buckling**, poor fire resistance (thermal conductivity), skilled work needed, no codes in many cases, **heat gain** in the roof

4. Gypsum Boards

Gypsum ($CaSO_4.2H_2O$) is a mineral found in Rajasthan, J&K;, Tamil Nadu; and **by-product gypsum (phospho-gypsum, FGD gypsum)** from industries. When heated at about **150 degrees C** it becomes **plaster of Paris (hemi-hydrate)** that sets when mixed with water.

Gypsum reaction

$CaSO_4.2H_2O$ --(heat 150 C)--> $CaSO_4.0.5H_2O + 1.5\ H_2O$ (calcination)

$CaSO_4.0.5H_2O + 1.5\ H_2O$ -> $CaSO_4.2H_2O$ (setting, in 8 to 15 minutes)

Gypsum product	Description / use
Gypsum plaster board (paper-faced)	Core of gypsum between two **paper liners**, thickness **9.5, 12.5, 15 mm**, size 1200 x 2400 mm; **ceilings and partitions** (dry wall); fixed on **GI metal studs**
Fibrous gypsum board / glass fibre reinforced gypsum (GFRG - "Rapidwall")	Panels with glass fibre in gypsum, **core filled with concrete and reinforcement** - load bearing walls and floors in mass housing
Gypsum plaster	Single coat **internal plaster 10 mm**, smooth, no curing, **saves 40-50% time and 20-30% cost** over cement plaster
Gypsum blocks and tiles	Light partition blocks, **false ceiling tiles**
Moisture resistant and fire resistant boards	Special additives for bathrooms and fire ratings (1-2 hours)
Cornices, mouldings	Decorative

- **Merits: light** (25 kg/m^2 for partition), **fast dry construction**, good finish, **fire resistant**, **thermal and acoustic insulation**, no curing water, uses **waste gypsum**; reduces dead load (good for earthquake)
- **Demerits: not for wet areas without treatment** (loses strength in water), low impact resistance, **non-structural** (boards), the metal frame needs protection; **skilled fixing**, imported systems costly

5. Timber Substitutions

Timber is costly and scarce because of **deforestation**. Doors, windows, frames, roof trusses and furniture can use alternative materials.

Substitute	Notes
Bamboo	Fast-growing, **strength comparable with steel by weight**; treated by **borax-boric acid** or smoke; used for **roof trusses, walls (ekra), scaffolds, bamboo reinforcement, flooring**; Bamboo mat boards and **bamboo mat corrugated sheets (BMCS)**
Plantation timber and treated softwood	Eucalyptus, rubber wood, poplar, **preservative treated** (CCA, boron)
Particle board, MDF, plywood (plantation), block board	Made from wood chips and **agri-residues** with resin

Substitute	Notes
RCC / precast concrete door and window frames, lintels	Durable, termite proof; no polishing
Steel and aluminium sections / frames	Rolled sections, pressed steel frames; **PVC and FRP**
Ferrocement / fibre-cement doors	Thin shutters, cheap
FRP (GRP), PVC / WPC (wood plastic composites)	Rot and termite proof, light
Jute-polymer composites, coir boards, wood-wool cement boards	Panels for partition, ceiling

- **Design economy:** use of **smaller sections**, **built-up sections, composite trusses**, **doors without frame (the frameless door)**, **sliding windows**, reduce area of shutters, planning for **modular door and window** sizes; **use RCC lintel-cum-sunshade**

6. Industrial Wastes in Building Materials

Industry produces millions of tonnes of **solid wastes** which pollute land, water and air. Using them in construction **saves cost and natural resources**.

Waste	Source	Use in construction
Fly ash	Thermal power plant (above 200 million t/yr in India)	**PPC cement, fly ash bricks, blocks, concrete admixture, lime-pozzolana, road embankments, light weight aggregate, FaL-G, cellular concrete**
Blast furnace slag (GGBS)	Steel plant	**Slag cement (PSC), slag bricks, aggregate,** geopolymer
Steel slag	Steel plant	Road base, aggregate
Phospho-gypsum, FGD gypsum	Fertiliser / power plants	**Gypsum plaster, blocks, boards, cement retarder**
Red mud	Aluminium (bauxite) industry	**Bricks, tiles, red mud-polymer composite doors / windows (BMTPC),** cement
Silica fume	Silicon metal industry	High performance concrete
Mine tailings, marble slurry, stone dust	Mining and stone cutting	**Bricks, blocks, M-sand, fillers**
Construction and demolition (C&D;) waste	Demolished buildings	**Recycled aggregate, bricks, paver blocks, non-structural concrete**

Waste	Source	Use in construction
Waste plastics, tyre rubber	Domestic / industrial	Plastic roads, tiles, rubberised concrete, lightweight panels
Spent foundry sand, ceramic waste, glass cullet, bagasse ash	Industry	Blocks, tiles, aggregates

Benefits of using industrial wastes

* Cuts cost and consumption of **cement, sand, clay** (conserves top soil and energy)
* **Solves disposal problem**, reduces pollution and landfill needs
* Many wastes improve the **durability** (low heat of hydration, resist sulphates)
* **Cautions:** quality control, **leaching of heavy metals**, radioactivity checks; code compliance

7. Agricultural Wastes in Building Materials

India produces **about 500 million tonnes of agricultural residue** each year; much is burnt (air pollution). They are **renewable and cheap**, and have low embodied energy.

Waste	Products
Rice husk / rice husk ash (RHA)	RHA has **85–90% amorphous silica** – a pozzolana: **lime–RHA cement, RHA concrete, bricks, blocks, boards, insulation**
Sugarcane bagasse and bagasse ash	**Particle boards, bagasse-cement boards, ash in concrete, fibre**
Wheat / paddy straw	**Straw bales** (walling), **straw boards (Stramit)**, mud-straw plaster, **thatch roof**
Coconut husk and coir	**Coir fibre boards, coir-cement, mats, ceilings**
Groundnut shell, cotton stalk, jute stick, sawdust, wood waste	**Particle boards, wood-wool cement boards, fibre boards**
Bamboo, reeds, elephant grass	Roof, wall panels, bamboo reinforced concrete
Fibres: sisal, jute, banana	**Fibre reinforced mud blocks, mortar, roofing sheets**
Palm leaves, areca nut husk	Thatch, boards

- **Problems: moisture absorption, rot, fire, insects** – need preservative treatment, coating, cement / lime / resin binders, and **proper detailing** (raised plinth, wide eaves)
- **Fire retardant treatment of thatch:** dipping in solutions of **borax + boric acid, ammonium phosphate / sulphate, or alum**; or covering the thatch with **mud–lime slurry plaster**

8. Alternative Building Maintenance

Maintenance is the **care** that keeps a building safe and usable, avoiding expensive repair. **Low cost housing** needs **simple, cheap and community-based maintenance**, with alternative materials.

- **Preventive maintenance** is better than repair: annual inspection of roof, drains, plaster, doors; keep **roof and surface drainage** clear
- **Cracks:** repair with **cement slurry, lime-surkhi mortar, epoxy injection**, stitching with bars; **damp proofing** using **lime-based / bitumen** coats, **integral waterproofing compounds** and **brick-bat coba waterproofing** for roofs
- **Termite treatment** (borax, chemical barrier) and **wood preservatives**; **lime wash** with salt / alum in rural houses every year; **cement-free paints (lime, clay, distemper)**
- **Mud houses:** re-plaster with **mud + cow dung + lime** after every monsoon; **thatch** re-laid every 3-5 years
- **Use of community / mason training**, **Building Centres** provide material, advice
- **Durable material choice** (ferrocement, FaL-G, stone) reduces cost over life: **life cycle cost approach**; **solar passive design** reduces repair of heat cracks

9. Low Cost Infrastructure Services

A house is useful only when it has **water supply, sanitation, drainage, roads, power and waste disposal**. The cost of infrastructure is **15-30%** of the cost of housing for the poor, so low cost technology is essential here too.

Present status

- **Water:** a large share of rural households have no tap water at home; **Jal Jeevan Mission (2019)** aims at **tap water connection to every rural household**; **AMRUT** in cities; slums depend on stand posts, tankers, hand pumps
- **Sanitation: Swachh Bharat Mission (2014)** has built millions of toilets and declared India ODF (2019); but **sewerage covers only about 30-35% of urban population**; septic tanks and pit latrines are widespread; **faecal sludge management** is a new challenge

- **Solid waste and drainage:** poor collection, open drains; **Energy:** electricity (**Saubhagya**), LPG (**Ujjwala**), biomass still used by many; **roads: PMGSY** for rural roads

10. Technological Options and Low Cost Sanitation

Low cost sanitation options

Option	Features
Pour-flush (water seal) latrine with twin pit (Sulabh)	**Water seal trap** (2-3 L flush) stops smell and flies; **two leach pits**, used alternately; **compost-like** safe sludge after 2 years; **cost Rs 12,000-15,000**
Single pit latrine, VIP (ventilated improved pit)	Dry pit with vent pipe, for no-water areas
Septic tank + soak pit	For houses with a bigger water use; **tank holds sewage for 24 h**, with 2 chambers; effluent to **soak pit**; desludging every 2-3 years
Community / public toilets	In slums with **bio-digester**, shared maintenance; sanitary complex
Eco-sanitation (ecosan), composting toilet	Urine diverting, **reuse as manure**; waterless
Biogas-linked toilet (bio-digesters, DRDO bio-toilet)	Anaerobic digestion, gas for cooking, **Bio-toilet** in trains
Simplified / condominial sewerage, small bore sewer	Low cost sewer for dense area, **shallow depth, small diameter**
Decentralised wastewater treatment (DEWATS), constructed wetlands	Sustainable local treatment

Design of twin leach pits (approximate)

Pit capacity = N x R x T (N = users, R = sludge accumulation rate = 0.04 m^3/person/year (wet cleansing) or 0.03 (dry), T = pit filling time = 2 to 3 years (each pit is used alternately))

For 5 users and T = 3 yr: V = 5 x 0.04 x 3 = 0.6 m^3; adopt a circular pit of **1.0 m diameter x 1.0 m depth (0.78 m^3)**; lined with **honeycomb brickwork, cover slab, 1.0 m apart** (centre to centre, double the pit dia), at least **10 m away from a water source**.

11. Domestic Waste (Wastewater and Solid Waste)

- **Grey water** (bath, kitchen, laundry) is **60–70%** of domestic wastewater; manage by **soak pits, kitchen garden reuse, trench drains, leach pits, and root-zone treatment**
- **Drainage: open surface drains (U-shaped, lined with brick, stone, or precast cement concrete channels) with gradient 1 in 200 or steeper**, covered with slabs; **lanes with central or side drains**; **storm water drains** separate; **no stagnation** to prevent mosquito breeding
- **Solid waste: segregation at source (wet / dry), composting (pit, windrow), vermicomposting, biogas plants, recycling, community bins** and door-to-door collection (**SWM Rules 2016**); **sanitary landfill** for the residue
- **Soak pit design:** capacity for the peak flow, e.g., 135 lpcd x 5 persons x 0.3 (grey part) = about 200 L/day => pit 1 m dia x 1 m deep filled with **brick-bats / stones**, with 300 mm sand layer; **not near wells**

12. Water Supply (Low Cost Options)

Option	Details
Hand pump (India Mark II / III)	Bore well with hand pump for **250 persons per pump**, up to 50 m depth
Public stand posts	Tap, 1 tap per **100–150 persons**, in slums (40 lpcd)
Gravity flow scheme / piped scheme	From a spring or lake in hills; **cheap, no power**
Open dug well and protected well	Shallow wells with parapet, platform, cover; **chlorinated**
Rain water harvesting (RWH)	Roof catchment, **first-flush** device, filter, **storage tank or recharge pit**; **V = A x R x C** (A = roof area, R = rainfall, C = 0.8)
Household treatment	**Boiling, bleaching powder, chlorine tablets, candle (ceramic) filters, slow sand filter, solar disinfection (SODIS), bio-sand filter**
Community RO / water ATM, tankers	Urban slums

Example: rain water harvesting potential
Roof area 50 m^2, annual rainfall 700 mm, run-off coefficient 0.8. V = 50 x 0.7 x 0.8 = **28 m^3** = 28,000 litres per year (about 77 L/day)

13. Energy

- **Energy in housing:** cooking (largest in rural houses), lighting, heating / cooling, water heating, appliances; low cost solutions **cut both bills and pollution**
- **Cooking: improved smokeless chulha (cookstove)** with 25-30% efficiency vs 10% of traditional one; **biogas plant (family size 2-4 m^3 from cattle dung)**, **LPG (Ujjwala)**, **solar cookers**
- **Lighting: LED lamps (UJALA)**, **solar lanterns and home lighting systems, rooftop PV (PM Surya Ghar)**, **solar street lights**
- **Water heating: solar water heaters** (100-200 L) save electricity
- **Passive solar design / energy efficient buildings:** orientation, **cross ventilation, shading by chajjas, roof insulation (white paint, filler slab, mud phuska), thick walls and cavity / rat-trap walls, light colours, skylights**, ECBC (Energy Conservation Building Code)
- **Others: wind mill and micro hydel** in suitable areas, **community biogas, bio-mass gasifiers, solar pumps**

14. More Alternative Materials at a Glance

Material	Key facts
Stabilised soil (compressed earth) blocks – CSEB	Soil + 5–10% cement (or lime) pressed in a manual / hydraulic press (**Auram press**); size 230 x 190 x 100 mm; **strength 3–5 MPa**; no burning; **30–40% cheaper than brick**; thermal comfort
Autoclaved aerated concrete (AAC) blocks	Fly ash + lime + cement + aluminium powder, autoclaved; **density 550–650 kg/m^3**, light, insulation, fast laying with thin-joint adhesive
Cellular light weight concrete (CLC)	Foam concrete, density 400–1800 kg/m^3, panels and blocks
Micro concrete roofing (MCR) tiles	Thin cement-sand tiles 6–12 mm, **cheap roof for rural houses**, made in small units; **cost 30% less than clay tiles**
Bamboo reinforced concrete	Treated bamboo strips as steel substitute in **lintels, slabs, columns** with low loads; **bond improved with bitumen / sand coat**
Fibre cement / asbestos-free sheets, bamboo mat corrugated sheets	Light roofing
Sandwich (EPS) panels / PUF panels	Two cement skins with insulating core; quick walls
Geopolymer concrete	Fly ash + alkali activators, **no cement**, low CO_2
Plastic / recycled plastic bricks and tiles	Waste plastic bonded, low cost

- Before use, a new material must pass **tests of strength, durability, water absorption and fire**; get **BIS or BMTPC (PACS) approval**; pilot demonstration; **build skill in the community** through training at **Building Centres**

Remember for exam

* Ferrocement: thin (10–40 mm), mortar 1:2 with multiple wire mesh layers, 2–8% steel; cure 14 days
* Gypsum board: paper-faced boards on metal studs; GFRG panels for load bearing; not for wet areas
* Fly ash, slag, red mud, phospho-gypsum = industrial wastes; RHA, bagasse, straw, coir = agricultural wastes
* Bamboo, ferrocement, RCC, FRP, particle board replace timber

* Twin pit pour-flush latrine, 1 m dia pits; hand pump for 250 persons; improved chulha, biogas, solar

Important questions

1. Suggest substitutes for scarce building materials such as cement, steel and timber.
2. What is ferrocement? Explain its materials, construction and applications.
3. Write a note on gypsum boards and their use in low cost housing.
4. Explain the alternatives to timber in doors, windows and roofs.
5. Discuss the use of industrial wastes and agricultural wastes as building materials.
6. Explain alternative building maintenance for low cost houses.
7. Describe the present status and technological options of low cost infrastructure services.
8. Explain low cost sanitation options. Design a twin pit latrine.
9. Discuss low cost water supply and energy options for housing.